

# Paraphenylenediamine



## What is it?

Paraphenylenediamine (PPD) is the main pigment in hair dye. It is found in most permanent, demi permanent and two-part (two bottle) hair dyes. It is in some tints and rinses.

Allergy to hair dyes can produce swelling, blistering, or scaling. It may show up as swelling of the eyelids and ears, or as a rash on the forehead, neck, or side of the face. In general, once the dye has “set,” it is non allergenic. However, sometimes not all the dye sets and dyed hair can continue to cause allergy.

PPD is sometimes used to dye other products, such as furs and cloths. PPD can also be found in temporary black henna tattoos. PPD can also be found in black rubber products (i.e. black rubber hoses, tires, and tool grips).

Pure natural henna should not contain PPD. Some individuals with PPD allergy can use hair dyes with para toluenediamine sulfate (PTDS).

## PPD can also be known as:

- 4-phenylenediamine
- 1,4-diaminobenzene
- 1,4-benzenediamine
- 4-aminoaniline
- p-diaminobenzene

## Some Possible Uses of PPD / Where it might be found:

- Hair dye
- Comb-in hair tint and hair rinses
- Henna tattoo (black henna)
- Shampoo-in highlight, lowlights
- Skin paint, dark makeup, dark lipstick
- Dye for socks, support hose, shoes

- Textile, rubber, and fur dyes
- Violin chin rest stain
- Antioxidant in antifreeze, fuels, and rubber
- Corrosion inhibitor in oils
- Gas sweetener
- Plastic manufacture
- Rubber antioxidant
- Printing ink
- Anti-ozonant
- Milk and water testing reagent
- Retarder in acrylate production
- Lithography and photocopying
- Photo or x-ray film developer

## Other products that may cross react to PPD:

- N-isopropyl-n-phenyl-p-phenylenediamine
- p-toluenediamine
- p-aminodiphenyl-amine
- 2,4-diaminoanisole
- o-aminophenol
- P-aminoazobenzene
- Paratoluene diamine (PTDS)
- 2-Methoxymethyl-p-phenylenediamine

Other para amino compounds, such as parabens, benzocaine anesthetics, para aminobenzoic acid (PABA), azo / disperse dyes, 4,4-diamino diphenylmethane, thiurams, aminoazobenzene

Reactions can occur from taking related products such as saccharin sweeteners, thiazides, sulfonamides, sulfonylurea anti diabetics, PAS, or celecoxib. Let your doctor know if this happens.

## What should I do to avoid PPD?

Do not use products that contain PPD. Check the complete ingredient list of everything you use. For

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products already at home that do not list ingredients, go to the store and inspect the original box or package. Some products' ingredients are found on the internet at the manufacturer's or store's website.

Tell your beautician, hairdresser, barber, medical provider, nurse, and others to avoid using products that contain PPD in your care.

To find a possible workplace exposure, check the Material Safety Data Sheet (MSDS). Check the complete ingredient list of each product you encounter. Wear long rubber or vinyl gloves to protect against contact with PPD at work.

Always check labels to make sure the product you are using is PPD-free. Not every product from a specific company will be PPD and/or PTDS free.

Some of hair dyes that may be PPD-free:

- PPD and PTDS-free hair dyes: All Semi Permanent: Arctic Fox, Creme of Nature, Good Dye Young, Got2b, L'Oreal Paris, Lime Crime, Manic Panic, Matrix, Wella
- Temporary rinses: These use pigment dyes that wash out.
- Gray: Grecian Formula 16 liquid with conditioner, Grecian Plus gradual haircolor foam
- Plant Based Dyes: Pure henna and indigo, vegetable hair dye. Options include: Hannah Natural, It's Pure Organics, Lush Henna Hair Dye, Desert Shadow Hair Colour, Light Mountain, Surya Brasil, Herbatint Vegetal, Saach Organics Natural Hair Colour, The Henna Guys, Vegetal

- PPD-free dye that contains PTDS: Wella Color Charm Demi Permanent, Schwarzkopf Keratin Color, Sanotint Light, Madison Reed Permanent and Demi-Permanent
- Some Goldwell brands may be PPD-free

In case of a severe reaction to PPD dye, wash your hair and scalp well with a mild shampoo or soapless cleaner. Then, apply a compress of old cotton cloths moistened with 2% hydrogen peroxide solution or a 1:5000 dilution of potassium permanganate to completely oxidize the dye. Once PPD is fully developed or oxidized, it no longer causes an allergic response. You may apply cool olive oil or lime to help ease tightness and soften crusts.

*In addition, the ACDS Contact Allergen Management Program® (CAMP®) can help manage contact dermatitis. It helps to find products that are safe to use by creating a list of products that do not contain one's allergens. For more information about CAMP®, please ask your doctor or visit [www.contactderm.org](http://www.contactderm.org).*

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